

Q1). Design a Class Hierarchy

class Person

name, age

input Person()

Read name

Read age

End class

class Student inherits Person

regNO, marks[5], avg

calculate Avarage()

Sum = 0

For i = 0 to 4

Read marks[i]

Sum = Sum + marks[i]

avg = Sum / 5

End class

class ResearchStudent inherits Student

research Area

display complete Details()

Print name, age

Print regNO

Print avg

Print research Area

End class.

main

create object rs of ResearchStudent

rs.input Person()

Read rs.regNO

rs.calculate Avarage()

Read rs.researchArea

rs.display complete Details()

End main

Q2) Complete the missing Pseudocode

BEGIN

READ N

READ ARRAY A

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow i+1$ TO $N-1$

IF $A[i] == A[j]$

$A[j] = -1$

END IF

END FOR

END FOR

PRINT modified ARRAY

END .

Q3). Debug the Psuedocode.

Class Area

METHOD area(length)

RETURN length * length

END METHOD

METHOD area(length, breadth)

RETURN length * breadth

END METHOD

END CLASS

Q4) Trace Output: -

(i) Operation

Sum [value]

1

0-1

2

-1+2

3

• 1+3

4

-2+4

5

2-5

Final Output

-3

Q5) write Efficient Pseudocode.

Begin

READ N

READ ARRAY A

First $\leftarrow -\infty$

second $\leftarrow -\infty$

FOR $i \leftarrow 0$ TO $N-1$

IF $A[i] > \text{First}$

second $\leftarrow \text{First}$

First $\leftarrow A[i]$

ELSE IF $A[i] > \text{second}$ AND $A[i] \neq \text{First}$

Then

second $\leftarrow A[i]$

ENDIF

END FOR

Print second

END

Q6) Convert Problem Statement into pseudocode:-

Inter FACE Payment

method pay(amt)

END interface

CLASS CreditCard Implements Payment

method pay(amt)

END method

END class

CLASS UPI Implements Payment

method pay(amt)

END method

END CLASS

CLASS Net banking Implements Payment

method pay(amt)

END method

END class

BEGIN

READ choice, amt

IF choice = 1

P ← creditCard

ELSE IF choice = 2

P ← UPI

ELSE

P ← Net banking

END IF

P.pay(amt)

END

Q1). Fill in the blanks.

Class Dog inherits Animal

Animal Obj $\leftarrow$ new dog()

Q8).

Correct constructor.

CLASS Employee

DATA

id

name

constructor (id, name)

this.id $\leftarrow$ id

this.name $\leftarrow$ name

END constructor

END class.

Q9) Case Study (Integrated problem)

class Vehicle

VehicleNO, Owner name

METHOD display()

END class

CLASS Car inherits Vehicle

Fuel type

ENDCLASS

CLASS Truck inherits Vehicle

load Capacity

END class

BEGIN

READ N

FOR $i \in 1 \text{ TO } N$

READ type

IF type = "car"

Obj $\leftarrow$ NEW Car

ELSE

Obj $\leftarrow$ NEW Truck

ENDIF

Store Obj

ENDFOR

FOR each Obj

Obj.display()

ENDFOR

END